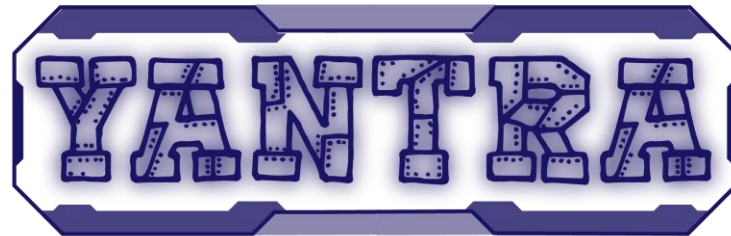


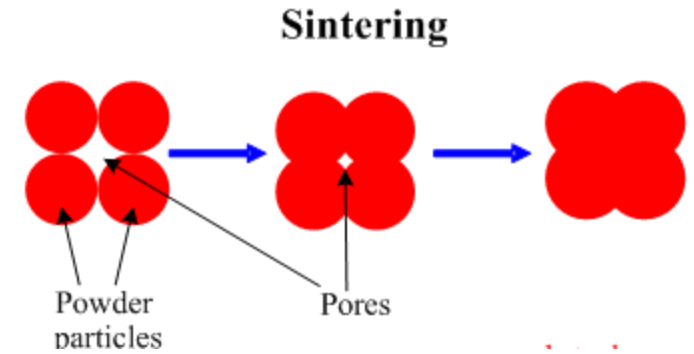
Selective laser sintering



Selective laser sintering

Selective laser sintering (SLS) is a free-form fabrication method based on **sintering** of the metallic or nonmetallic powders selectively into an individual object.

a thin layer of **fusible powder** is laid down and heated to just **below its melting point** by **infrared heating** panels at the side of the chamber.



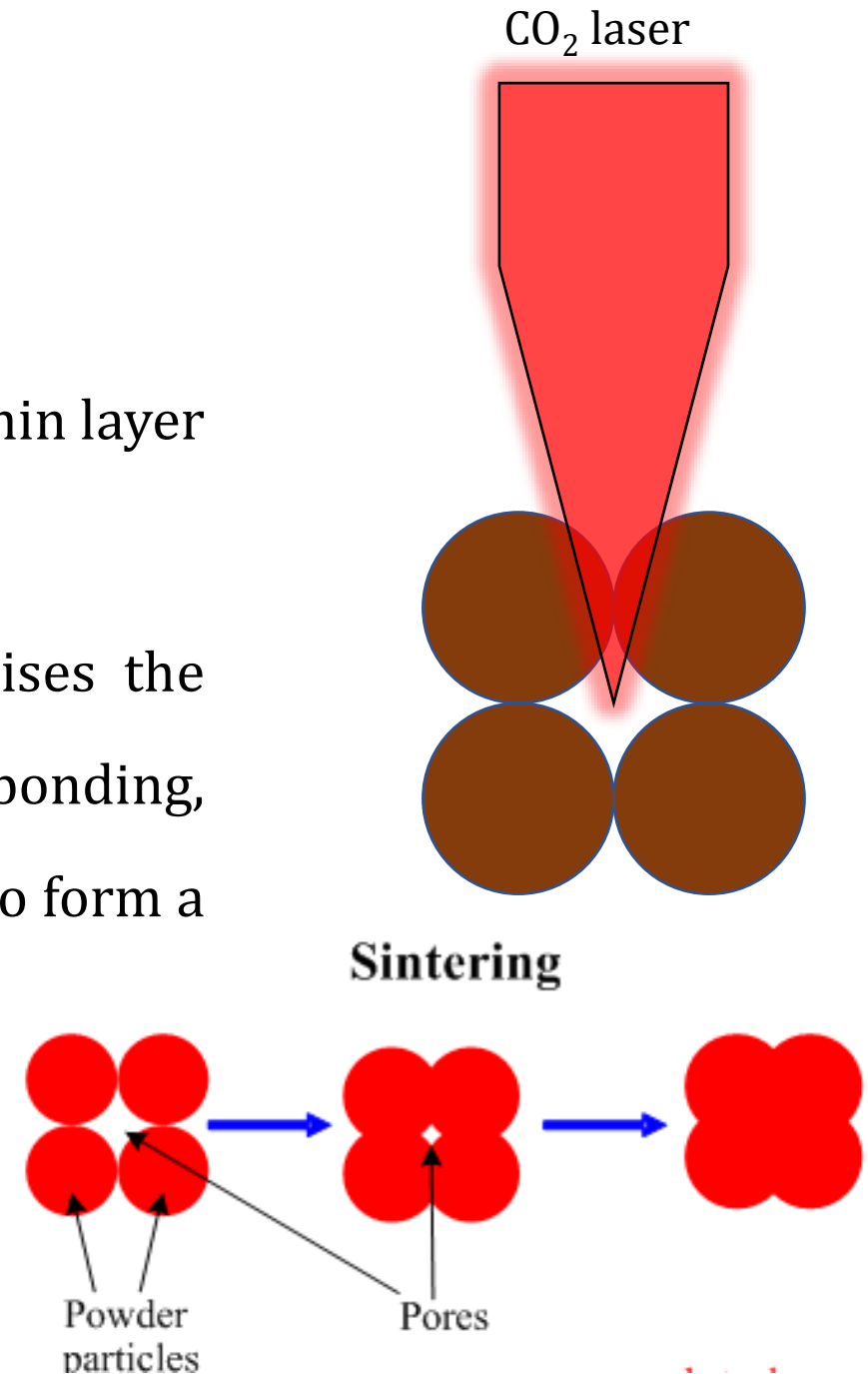
Selective laser sintering

SLS processes were originally developed to produce plastic prototypes using a point-wise laser scanning technique.

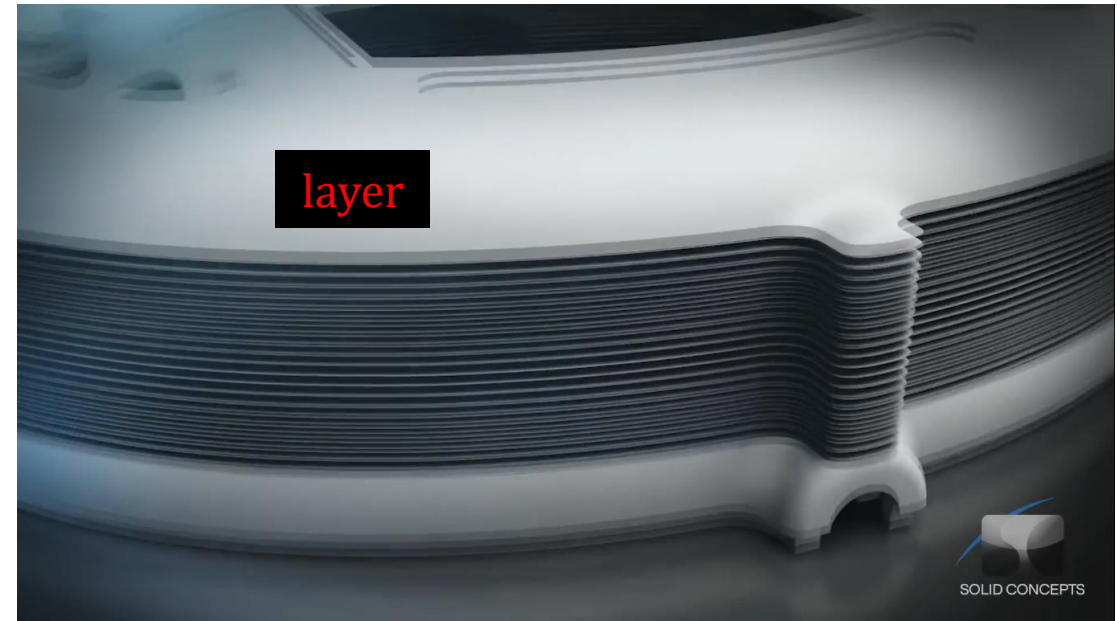
This approach was subsequently extended to metal and ceramic powders; additional thermal sources are now utilized; and variants for layer-wise fusion of powdered materials are being commercially introduced.

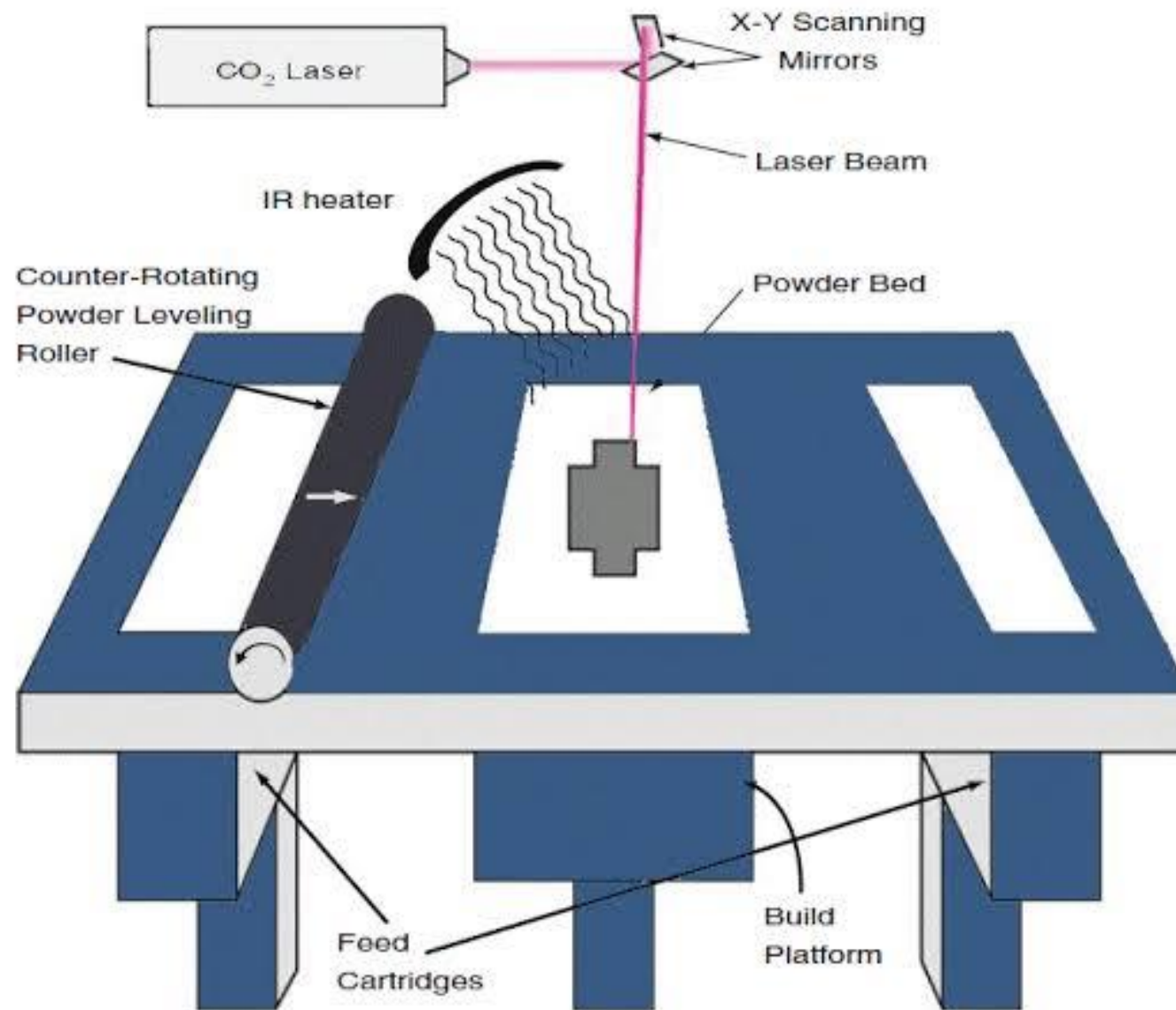
PRINCIPLE

- The SLS process is based on the following two principles:
 1. Parts are built by sintering when a **CO₂ laser beam** hits a thin layer of **powdered material**.
 2. The interaction of the laser beam with the powder raises the temperature to the point of melting, resulting in particle bonding, fusing the particles to themselves and the previous layer to form a solid.



- The building of the part is done **layer by layer**.
- Each layer of the building process contains the cross-sections of one or many parts.
- The next layer is then built directly on top of the sintered layer after an additional layer of powder is deposited via a roller mechanism on top of the previously formed layer.





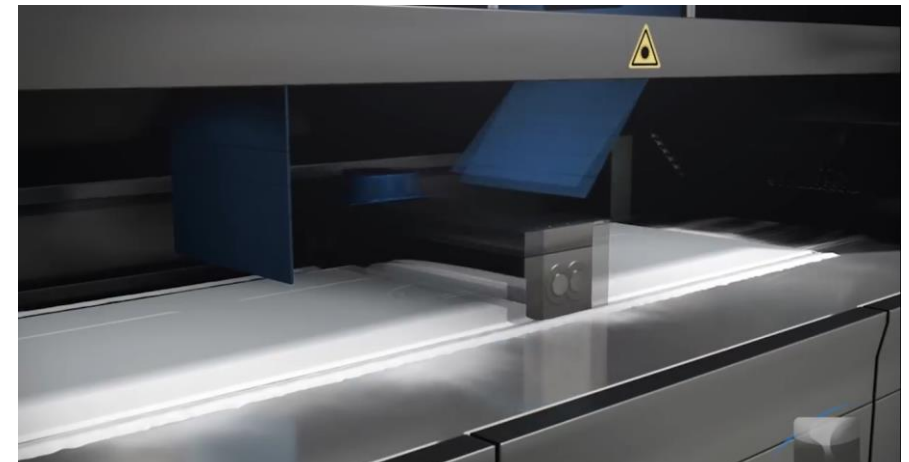
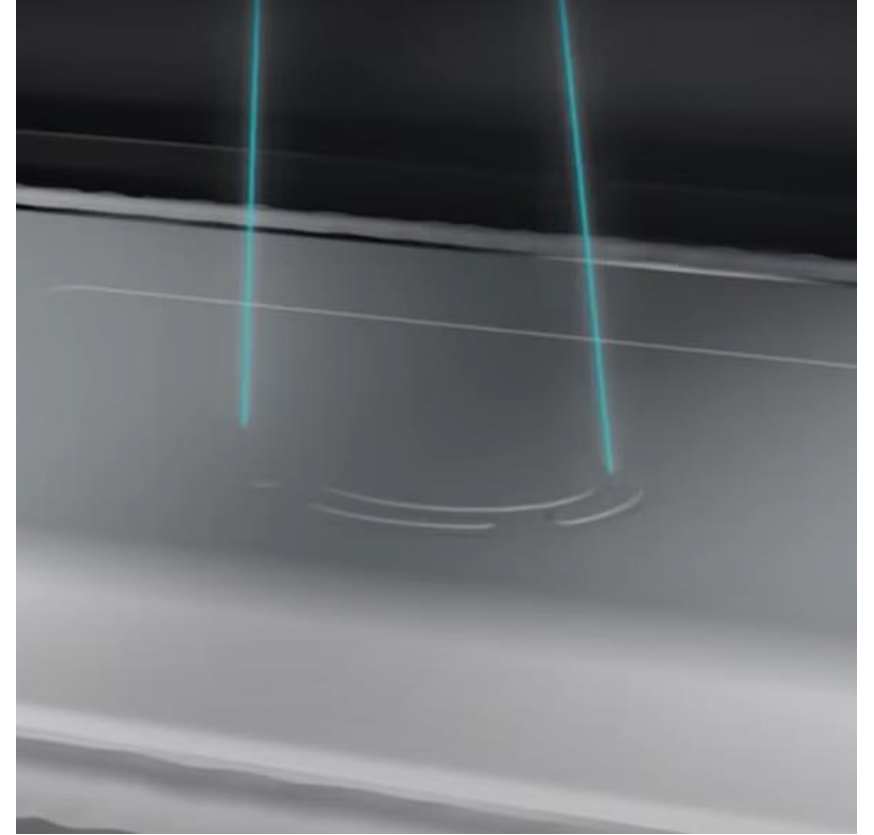
Process

The part building process takes place inside an enclosed chamber filled with nitrogen gas to minimize oxidation and degradation of the powdered material.

The powder in the build platform is maintained at an elevated temperature just below the melting point of the powdered material.

Infrared heaters are placed above the build platform to maintain an elevated temperature around the part being formed, as well as above the feed cartridges

In some cases, the build platform is also heated using resistive heaters



roller spreads out another layer of powder

Process

Once an appropriate powder layer has been formed and preheated, a focused CO2 laser beam is directed onto the powder bed and is moved using galvanometers in such a way that it thermally fuses the material to form the slice cross section.

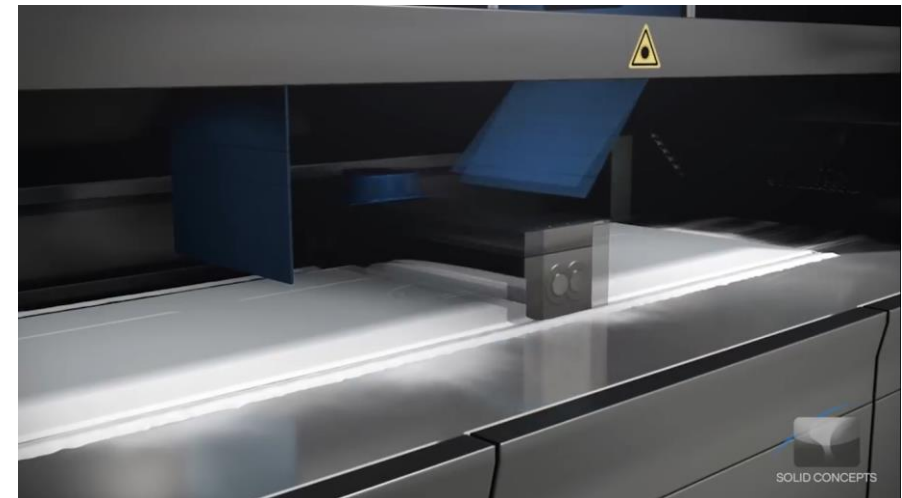
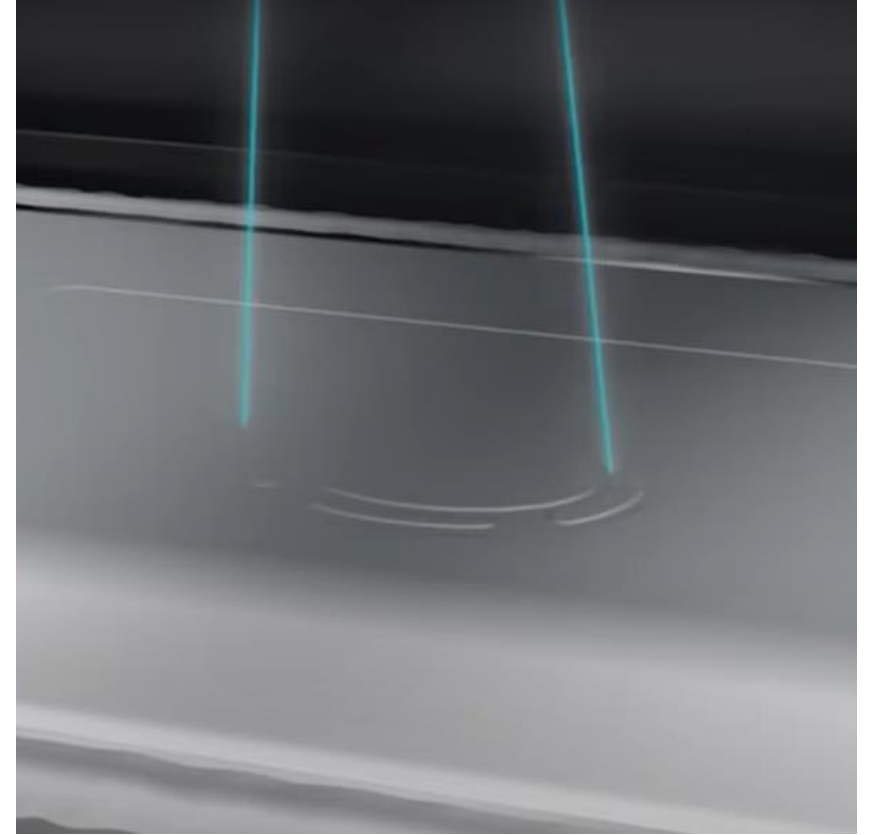
Surrounding powder remains loose and serves as support for subsequent layers, thus eliminating the need for the secondary supports

After completing a layer, the build platform is

lowered by one layer thickness and a new layer of powder is laid and leveled

using the counter-rotating roller. The beam scans the subsequent slice cross section.

This process repeats until the complete part is built.



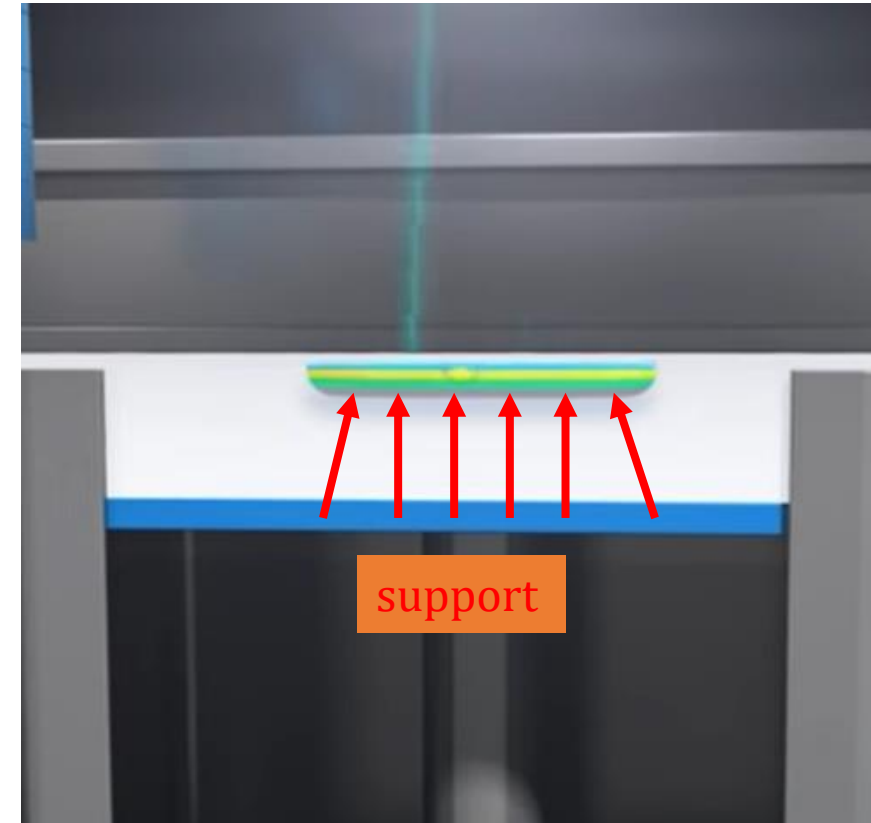
roller spreads out another layer of powder

SLS relies on the **supporting strength** of the **powder** around the partially fused object.

Therefore, **support columns** for any overhanging parts of the component are **not needed**.

A **variety** of materials can be used including polymers, epoxy, wax, metals, and ceramic with the appropriate binder.

After the object is fully formed, the piston is raised to **elevate it**.



Excess powder is simply brushed away
and a final manual finishing may be carried out.

Although **surface finishes** and **accuracy** are **not quite as good**
as with **stereolithography**,

The material properties can be quite close to those of the
intrinsic ones.

The method has also been extended to provide direct
fabrication of metal and ceramic objects and tools



surface finishes and accuracy are not quite as good

Materials for SLS

- 1. **Polyamide:**
- This material is used to create **rigid and rugged plastic** parts for functional engineering environments.
- This material is durable, can be **machined** or even **welded** where required.
- A variation of this material is the polyamide based composite system, incorporating glass-filled powders, to produce even more rugged engineering parts.
- This composite material improves **the resistance to heat and chemicals.**



2. Thermoplastic elastomer:

Flexible, rubber-like parts can be prototyped using the SLS.

the material produces parts with **high elongation**.

Yet, it is able to **resist abrasion** and **provides good part stability**.

The material is impermeable to water and ideal for sports shoe applications and engineering seals.



3. Polycarbonate:

- An industry-standard engineering thermoplastic.
- suitable for **creating concept** and functional models and prototypes, investment casting patterns for metal prototypes and cast tooling, masters for duplication processes, and sand casting patterns.
- These parts can be built quickly and are excellent for prototypes and patterns with fine features.



Nylon:

Another industry-standard engineering thermoplastic.

suitable for creating models and prototypes that can withstand and perform in demanding environment.

one of the **most durable** rapid prototyping materials currently available in the industry, and it offers substantial heat and chemical resistance.

A variation of this is the Fine Nylon and is used to create fine-featured parts for working prototypes. It is durable, resistant to heat and chemicals, and is excellent when fine detail is required.

5. Metal:

- polymer coated stainless steel powder is infiltrated with bronze.
- excellent for producing **core inserts** and preproduction tools for injection molding prototype polymer parts.
- exhibits high durability and thermal conductivity and can be used for relatively large-scale production tools.

6. **Ceramics**: these use zircon and silica coated with phenolic binder to produce complex sand cores and molds for prototype sand castings of metal parts.

PROCESS PARAMETERS

1. Part bed temperature:

The part bed is the **central region** of the SLS machine where the part is built.

The temperature is controlled primarily by the **heater** underneath the build area. The temperature should be **lower than melting temperature** of powder.

The **higher the temperature** is set, **the lesser the incident energy** required during the SLS process.



part bed

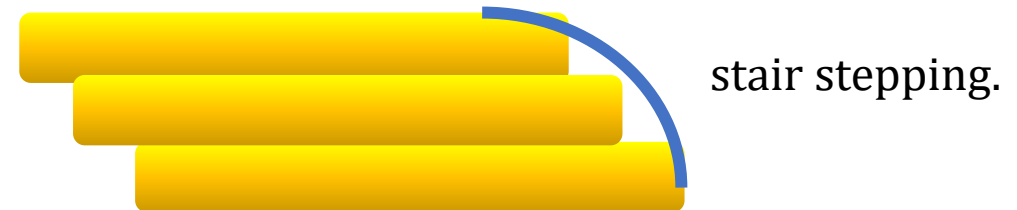
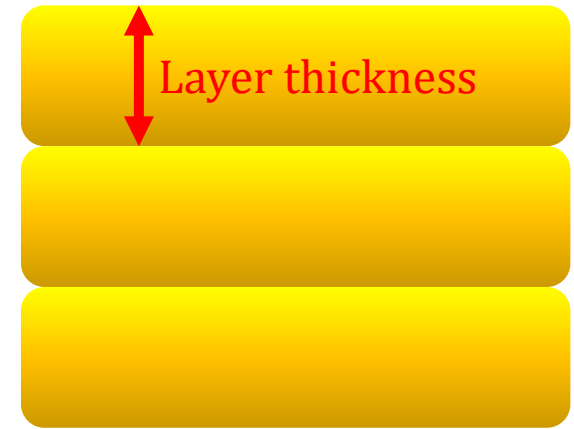
2. Layer thickness:

Layer thickness is the measure of the thickness of each layer by which the part is built during the SLS process.

It is also the depth by which the part piston is lowered after the laser scanning of each layer.

The **thicker the layer** requires **greater incident energy**.

But thicker layers will result in poor surface finish because of stair stepping.



3. Energy density:

defined as **the amount of energy input per unit area.**

It is dependent upon laser power, scan speed and scan spacing.

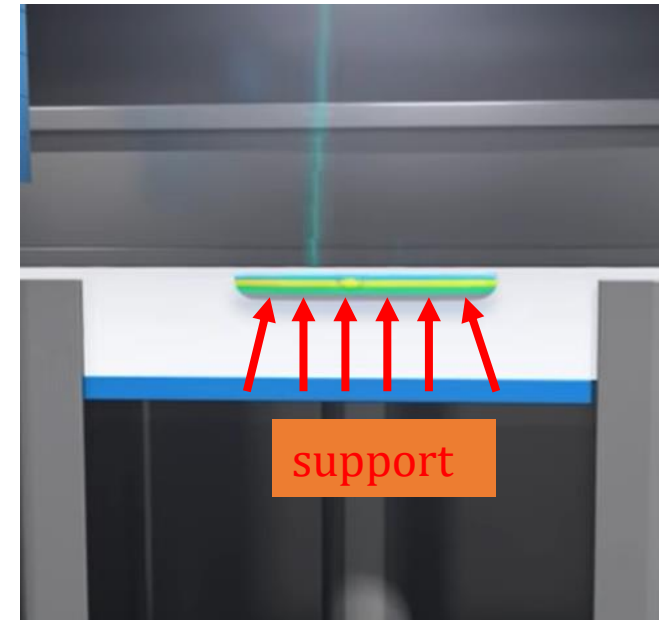
The laser power, scan speed and scan spacing need to be optimized according to the amount of input energy required to fuse the particles in the layer.

$$\text{Energy density} = \frac{\text{laser power}}{\text{scan speed} \times \text{scan spacing}}$$

Advantages of SLS

- Good part stability: The process and materials provide for directly produced **functional parts** to be built.
- Wide **range of processing materials**: thus providing flexibility and a wide scope of functional applications.
- It produces the **toughest components** compared to other RP techniques.
- A **relatively short production time** (approximately 1 in/h) is possible.
- **Little post-processing required**: The finishing of the part is reasonably fine and requires only minimal post-processing such as particle blasting and sanding.

- No post-curing required:
- No part supports required: The system does not require CAD developed support structures.
- This saves the time required for support structure building and removal. Leftover powder can be reused.
- Different parts can be built right on top of others. Multiple parts can be produced simultaneously.



Disadvantages of SLS

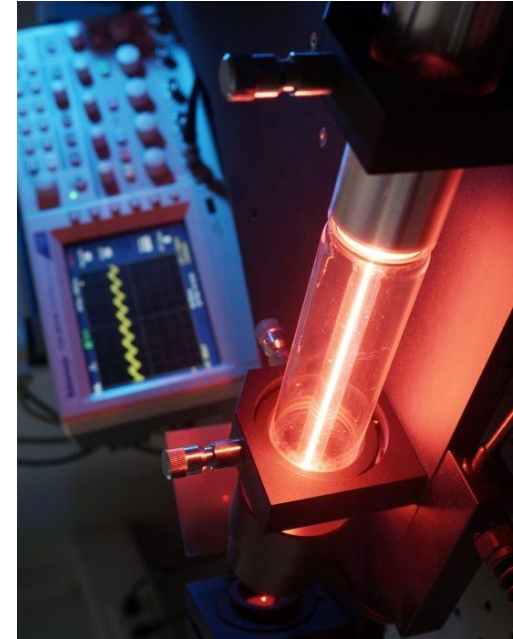
- A **rough surface** finish is produced due to the stair-step effect.
- The time needed to heat up the powder and cool down the model after it has been completed is approximately 8 to 10 h.
- **Large physical size of the unit:** The system requires a relatively large space to house it. Apart from this, additional storage space is required to house the inert gas tanks used for each build.



surface finishes and accuracy
are not quite as good



- **High power consumption:** The system requires high power consumption due to the high wattage of the laser required to sinter the powder particles together.
- **Toxic gases** are produced especially when plastic materials are fused.
- Components are **less detailed** than those produced by **stereolithography** due to a high **processing temperature** and **high shrinkage**.



Comparison
between SLS
and SLA



APPLICATIONS

- Concept models: Physical representations of designs used to review design ideas, form and style.
- Functional models and working prototypes: Parts that can withstand limited functional testing, or fit and operate within an assembly.
- Polycarbonate patterns: Patterns produced using polycarbonate, then cast in the metal of choice through the standard investment casting process. These build faster than wax patterns and are ideally suited for designs with thin walls and fine features. These patterns are also durable and heat resistant.
- Metal tools : Direct rapid prototype of tools of molds for small or short production runs.